

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Built and evolved **production software systems** serving millions of banking customers, taking **end-to-end ownership** from feature design through implementation, testing, and deployment using **Java Spring Boot, Python, REST APIs**, and **AWS** (EC2, S3, Lambda, CloudWatch); reduced incident resolution time by **35%** through improved monitoring and alerting.
- Integrated **AI and machine learning** into software development workflows, leveraging **Python data stack** (pandas, NumPy, scikit-learn) for anomaly detection and predictive maintenance in payment systems; implemented automated data pipelines reducing manual intervention by **60%**.
- Mastered diverse technology stacks across **microservices and monolithic architectures**, selecting and deploying **Docker, Kubernetes, gRPC**, and **REST** patterns for scalability; simplified legacy system complexity by refactoring **4 interconnected services** to reduce deployment time from **45 minutes to 12 minutes**.
- Simplified and improved critical payment-operations architectures by questioning every layer of complexity, eliminating technical debt in **PostgreSQL, MySQL**, and **Oracle** schemas while maintaining **GDPR** compliance and audit-trail integrity; delivered **ISO 20022 message flow** enhancements supporting cross-border and SEPA payment processing for enterprise banking clients.
- Demonstrated exceptional reasoning ability and drive by building **end-to-end reliability engineering systems**—designing **Splunk** and **Dynatrace** monitoring dashboards, **Jenkins CI/CD pipelines**, and **change-control processes** that reduced production incidents by **40%** and enabled autonomous team deployment ownership.
- Acted as a reliable, transparent team collaborator, supporting UAT coordination, post-go-live troubleshooting, and cross-functional integration with banking domain business analysts and operations teams; delivered **Power BI dashboards** and operational reporting that improved SLA visibility and enabled data-driven decision-making across interconnected enterprise business systems and ERP platforms.

SKILLS

Backend & Full-Stack Engineering – Java Spring Boot, Python, REST APIs, gRPC, microservices, monolithic systems, Flask, FastAPI.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, CI/CD (Jenkins, Git), Linux, shell scripting.

Data Engineering & BI – SQL (PostgreSQL, MySQL, Oracle), Python data pipelines (pandas, NumPy), Power BI, Tableau, DAX, data modelling, data warehousing, ETL.

Domain Expertise – SWIFT messaging (MT/MX formats), ISO 20022 message flows, SEPA payments, cross-border payments, PSD2, Open Banking, payments processing, ERP integration, change-control management, GDPR, audit-trail compliance.

Monitoring, Observability & Reliability – Splunk, Dynatrace, production support, SLA management, incident response (P1/P2), architectural simplification.

Analytics & Statistical Methods – Machine learning (supervised/unsupervised), hypothesis testing, A/B testing, time-series analysis, R, RStudio, scikit-learn.

Professional & Collaboration – Agile/Scrum, Jira, Confluence, UAT coordination, cross-functional stakeholder management, IELTS Academic Band 7/9.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

AI-Augmented Payment Transaction Validation & Routing Engine (HCL Technologies)

- Architected and delivered an **end-to-end production system** integrating **AI-powered machine learning models** (Python scikit-learn, pandas) into real-time payment message processing, combining **ISO 20022 message flow validation**, **SEPA compliance rules**, and predictive routing logic to eliminate manual intervention in cross-border payments.
- Implemented **microservices architecture** using **Java Spring Boot** REST APIs and **Python FastAPI** workers, deployed on **AWS (EC2, Lambda)** with **Docker containerization** and **Kubernetes orchestration**, achieving **99.95% uptime SLA** and processing **2.5M+ daily payment transactions** with sub-100ms latency.
- Built comprehensive **Splunk monitoring dashboards** and **CloudWatch alerts** for real-time transaction visibility, payment flow anomaly detection, and **GDPR audit-trail compliance**; reduced payment reconciliation errors by **73%** and incident detection time from **8 minutes to 45 seconds**.
- Simplified system architecture by refactoring legacy monolithic payment-validation logic into **5 stateless microservices**, eliminating **Oracle schema bottlenecks** through **PostgreSQL optimization** and async message queues; improved deployment velocity from **monthly releases to daily deployments**.
- Demonstrated relentless simplification by replacing complex conditional routing rules with an **ML-driven decision model** (80% accuracy) trained on historical payment data, reducing operational overhead and improving customer experience for enterprise banking clients across 12 jurisdictions.

Enterprise ERP Integration & Operational Analytics Dashboard (HCL Technologies)

- Designed and deployed a **Power BI enterprise dashboard** consolidating operational metrics across interconnected **ERP modules**, banking-core systems, and payment-operations platforms; delivered **real-time DAX measures** and **Power Query ETL pipelines** enabling business stakeholders to monitor transaction flow, SLA compliance, and **GDPR audit controls**.
- Executed **change-control management** for system configuration updates across **PostgreSQL**, **MySQL**, and **Oracle** databases, coordinating UAT and post-go-live support; ensured zero regulatory compliance drift through automated **SQL data-validation scripts** and **audit-trail verification**.
- Integrated multiple business systems via **REST APIs** and database-level data flows, simplifying cross-system dependencies and reducing data reconciliation errors by **55%**; improved analytics data freshness from weekly to **near real-time (5-minute refresh cycles)**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)